

PCMigration Reconciliation

v4.0.0 - Complete Public Release Guide

Evidence-first Windows migration auditing, selective repair, verification, and rollback.

Default-deny repair: every generated action begins with Approved=NO. Preview is the default. Source artifacts, destinations, identities, and hashes are independently re-authorized before any destination mutation.

Release	Execution host	Capture schema	License
4.0.0	64-bit Windows PowerShell 5.1	4.0	MIT

Designed for: experienced Windows users, administrators, and migration technicians comparing one source PC with one destination PC in the intended migrated user session.

Not designed as: disk imaging, blind profile cloning, credential transplantation, enterprise orchestration, or a guarantee that two computers become identical.

Keep the source PC and independent backups intact until the final destination verification is complete and the destination has remained stable.

1. What this release consolidates

Version 4.0.0 combines the complete v3.2.1 package, the v3.3 native registry-backup revision, the Windows PowerShell 5.1 HTML hotfix, and the final shortcut and Unicode corrections into one coherent package.

Area	Behavior in v4.0.0
Shortcut reconciliation	Compares launch target, arguments, and working directory instead of treating every .lnk binary hash difference as a migration gap.
Shortcut reporting	Adds a human-readable relative location and a complete source/destination reconciliation ledger.
Registry safety backup	Optional native HKCU\Software and HKLM\SOFTWARE binary snapshots plus narrow exports; broad snapshots are forensic-only.
Unicode paths	Preserves legal non-ASCII filenames including μ and keeps U+00B5 distinct from Greek μ.
PowerShell 5.1 HTML	Joins body fragments into one string before ConvertTo-Html -Body.
Package validation	Packaged hash/parser failures remain fatal; unrelated files warn unless strict mode is requested.
Protected state	Explicitly excludes credentials, protected security databases, caches, indexes, and machine-bound state.
Repair safety	Requires individual approval, source authorization, preview, risk gates, backups, and post-repair verification.

Why the package remains modular

Capture, comparison, repair, verification, and rollback have different permissions and risk profiles. Keeping them as separate scripts makes each phase reviewable, prevents accidental apply behavior during inventory, and allows the validator to verify every component independently.

Package components

Script	Role
Test-PCMigrationPackage-v4.0.0.ps1	Integrity, parser, and regression validation.
Capture-PCMigrationState-v4.0.0.ps1	Read-mostly source/destination evidence capture; source payload copying is opt-in.
Compare-PCMigrationState-v4.0.0.ps1	Offline non-mutating comparison and report/plan generation.
Invoke-PCMigrationRepair-v4.0.0.ps1	Preview/apply approved actions with source re-authorization and backups.
Verify-PCMigrationState-v4.0.0.ps1	Fresh destination capture and complete re-comparison.
Rollback-PCMigrationRepair-v4.0.0.ps1	Preview/apply file and narrow-registry rollback.
PCMigration.Common-v4.0.0.ps1	Shared implementation; do not run directly.
RegistryBackup-v4.0.0.ps1	Optional registry safety backup helper; do not run directly.

2. Requirements and preparation

- Windows 10 or Windows 11 on source and destination.
- 64-bit Windows PowerShell 5.1. PowerShell 7 is not the supported execution host.
- Run as the intended migrated user. Elevation is recommended for machine-wide evidence and required for administrative actions.
- Use trusted, encrypted, access-controlled storage for source captures and repair backups.
- Close applications whose settings matter before the source capture.
- Do not retire or alter the source PC until final verification is complete.

Step 1 - Validate the package

```
Set-ExecutionPolicy -Scope Process Bypass -Force
Get-ChildItem -LiteralPath $PWD -Filter '*.ps1' | Unblock-File
.\Test-PCMigrationPackage-v4.0.0.ps1
```

Expected result:

Package validation passed: PowerShell parser, SHA256 manifest, and v4.0.0 regression checks.

Use strict package-directory checking when desired:

```
.\Test-PCMigrationPackage-v4.0.0.ps1 -StrictPackageContents
```

Do not continue after a parser error, missing packaged file, or checksum mismatch. Unrelated files are warnings in normal mode so a harmless adjacent file does not invalidate verified package content.

Step 2 - Capture the source

```
.\Capture-PCMigrationState-v4.0.0.ps1 `
-OutputPath 'P:\PCMigration-Source-v4.0.0' `
-InventoryMode Standard `
-CaptureSettingsPayload `
-SearchSecureFileCandidates `
-StartStoppedInventoryServices
```

Optional portable roots:

```
-AdditionalPortableRoots 'D:\Tools','C:\Utilities','C:\PortableApps'
```

Optional forensic registry safety backup:

```
-CaptureRegistrySafetyBackup
```

For narrow targeted exports without broad binary hives:

```
-CaptureRegistrySafetyBackup -RegistryBackupTargetedOnly
```

3. Source review and controlled service handling

Review every collector before relying on the capture:

```
Import-Csv 'P:\PCMigration-Source-v4.0.0\Capture-Status.csv' |
Format-Table Collector,Status,Records,ElapsedSeconds,Message -AutoSize -Wrap
```

Status	Meaning
Success	Collector completed. A message may still identify expected exclusions or warnings.
Partial	Coverage is incomplete; comparison must not treat an unobserved item as absent.
Unavailable	Required platform/provider capability was unavailable.
Failed	Collector did not produce trustworthy coverage; investigate or recapture.

Review the shortcut inventory with readable locations:

```
Import-Csv 'P:\PCMigration-Source-v4.0.0\Applications\Shortcuts.csv' |
Sort-Object RelativeLocation |
Format-Table RelativeLocation,TargetPath,Arguments,TargetExists -AutoSize -Wrap
```

Controlled third-party service quiescing

If a verified third-party service owns a legitimate application settings database, its exact service name can be supplied with `-QuiesceServiceName``. The script stops it only around settings-file capture and restores it in a `finally`` block. Every transition is recorded in `Diagnostics\Service-State-Changes.csv``.

```
-QuiesceServiceName 'ExactVerifiedServiceName'
```

Never guess a service name. Defender, Kaspersky, firewall, Search, networking, credential, AppX, VSS, installer, and core Windows services are denylisted and will not be quiesced.

Registry safety snapshots

- Created only when `-CaptureRegistrySafetyBackup`` is supplied.
- Use native sequential `reg.exe save``; HKLM\SOFTWARE already includes WOW6432Node.
- Stored under `Registry\Raw``; never authorized as repair payloads.
- Contain sensitive application/security configuration and require protected storage.
- Never restore either broad hive wholesale onto the destination.

4. Destination capture and initial comparison

Step 3 - Capture the untouched destination

```
.\Capture-PCMigrationState-v4.0.0.ps1 `
-OutputPath 'C:\MigrationAudit\Destination-Before-v4.0.0' `
-InventoryMode Standard `
-SearchSecureFileCandidates `
-StartStoppedInventoryServices
```

Do not use -CaptureSettingsPayload or -CaptureRegistrySafetyBackup on the destination. Destination payloads are unnecessary for comparison and can create confusion about source authority.

Step 4 - Compare

```
.\Compare-PCMigrationState-v4.0.0.ps1 `
-SourceCapture 'P:\PCMigration-Source-v4.0.0' `
-DestinationCapture 'C:\MigrationAudit\Destination-Before-v4.0.0' `
-OutputPath 'C:\MigrationAudit\Reconciliation-Before-v4.0.0'

Start-Process 'C:\MigrationAudit\Reconciliation-Before-v4.0.0\Summary.html'
```

Review order

Order	Report	Question answered
1	Capture-Coverage.csv	Which categories are safely comparable?
2	Regression-Checks.csv	Did known critical checks operate?
3	Shortcut-Reconciliation.csv	Is each shortcut equivalent, missing, different, or destination-only?
4	Shortcut-Gaps.csv	Which shortcuts have real functional gaps?
5	Likely-Migration-Gaps.csv	What likely failed to migrate?
6	Missing-Applications.csv	Which application evidence is absent?
7	Application-Settings-Gaps.csv	Which configuration files/values differ?
8	Windows-State-Gaps.csv	Which Windows/user settings differ?
9	Manual-Secure-Actions.csv	What requires vendor/manual secure handling?
10	Repair-Plan.csv	Which reviewed actions are technically available?

5. Shortcut reconciliation - review before repair

Windows `.lnk` files often contain machine-specific binary metadata. Earlier releases could label a shortcut as different - and machine-wide shortcuts as high risk - even when the destination already contained a functionally identical shortcut. Version 4.0.0 separates binary identity from launch behavior.

FunctionalStatus	Meaning	Repair plan
Equivalent	Same normalized launch target, arguments, and working directory.	No action.
Missing	No shortcut exists at the same logical root and relative path.	Eligible only when a safe source payload exists.
Different	Shortcut exists, but launch behavior differs.	Review target and arguments before approval.
Destination-only	Present only on destination.	Informational; never removed.

Show the complete shortcut ledger:

```
Import-Csv 'C:\MigrationAudit\Reconciliation-Before-v4.0.0\Shortcut-Reconciliation.csv' |
Sort-Object FunctionalStatus,RelativeLocation |
Format-Table RelativeLocation,DestinationPresent,FunctionalStatus,BinaryStatus,Risk,SourceTarget -AutoSize -Wrap
```

Show only functional gaps:

```
Import-Csv 'C:\MigrationAudit\Reconciliation-Before-v4.0.0\Shortcut-Gaps.csv' |
Sort-Object RelativeLocation |
Format-Table RelativeLocation,Status,DestinationPresent,TargetPath,Arguments -AutoSize -Wrap
```

Rows in Shortcut-Equivalent-Binary-Differences.csv are informational. They are intentionally suppressed from Repair-Plan.csv because the destination shortcut is already present and functionally equivalent.

Relative path examples

Root	Example RelativeLocation
Current-user Start menu	Start Menu (current user)\Programs\Vendor\Tool.lnk
All-users Start menu	Start Menu (all users)\Programs\Vendor\Tool.lnk
Current-user desktop	Desktop (current user)\Tool.lnk
All-users desktop	Desktop (all users)\Tool.lnk
Current-user Startup	Startup (current user)\Agent.lnk

6. Install applications, then regenerate the plan

Application installation must precede settings and shortcut restoration. Install required applications with compatible versions and vendor-supported methods. Reinstall security products cleanly. Sign in to password managers normally. Use application-supported sync/export/import for protected or version-bound data.

Step 5 - Recapture after application installation

```
.\Capture-PCMigrationState-v4.0.0.ps1 `
-OutputPath 'C:\MigrationAudit\Destination-PostApps-v4.0.0' `
-InventoryMode Standard `
-SearchSecureFileCandidates `
-StartStoppedInventoryServices
```

Step 6 - Recompare and create a clean post-application plan

```
.\Compare-PCMigrationState-v4.0.0.ps1 `
-SourceCapture 'P:\PCMigration-Source-v4.0.0' `
-DestinationCapture 'C:\MigrationAudit\Destination-PostApps-v4.0.0' `
-OutputPath 'C:\MigrationAudit\Reconciliation-PostApps-v4.0.0'
```

Use the new post-application Shortcut-Reconciliation.csv and Repair-Plan.csv. Do not merge or reuse the pre-installation plan.

Review the plan

```
$PlanPath = 'C:\MigrationAudit\Reconciliation-PostApps-v4.0.0\Repair-Plan.csv'
$Plan = Import-Csv -LiteralPath $PlanPath
$Plan | Sort-Object {[int]$_ .Priority},Risk,Method,Item |
Format-Table ActionId,Approved,Priority,Method,Risk,RequiresAdmin,Item -AutoSize -Wrap
```

Approve only reviewed IDs:

```
$ApprovedActionIds = @('A00012','A00018')
foreach($Row in $Plan){
    $Row.Approved = if($Row.ActionId -in $ApprovedActionIds){'YES'}else{'N0'}
}
$Plan | Export-Csv -LiteralPath $PlanPath -NoTypeInformation -Encoding UTF8
```

Edit only Approved. Do not alter method, destination, artifact, identity, package, feature, capability, or hash fields.

7. Preview, apply, and verify

Preview low/medium current-user actions:

```
.\Invoke-PCMigrationRepair-v4.0.0.ps1 `
-SourceCapture 'P:\PCMigration-Source-v4.0.0' `
-PlanPath $PlanPath
```

Preview approved high-risk or administrative actions with their gates:

```
.\Invoke-PCMigrationRepair-v4.0.0.ps1 `
-SourceCapture 'P:\PCMigration-Source-v4.0.0' `
-PlanPath $PlanPath `
-AllowHighRisk `
-AllowAdminChanges
```

Apply only after the preview is correct:

```
.\Invoke-PCMigrationRepair-v4.0.0.ps1 `
-SourceCapture 'P:\PCMigration-Source-v4.0.0' `
-PlanPath $PlanPath `
-AllowHighRisk `
-AllowAdminChanges `
-Apply
```

The script writes `Approved-Repair-Plan.csv`, `Backup-Manifest.csv`, `Repair-Results.csv`, and file/registry backups to a timestamped destination Desktop folder. The backup manifest is updated before mutation so an interrupted run retains recovery evidence.

Sign out and sign in

A complete sign-out reloads Explorer, Console Host, regional/clock, font, language, personalization, and shell state. Restart Windows when an installer, feature, capability, or vendor application requires it.

Final verification

```
.\Verify-PCMigrationState-v4.0.0.ps1 `
-SourceCapture 'P:\PCMigration-Source-v4.0.0' `
-DestinationCapturePath 'C:\MigrationAudit\Destination-After-v4.0.0' `
-ReportPath 'C:\MigrationAudit\Reconciliation-After-v4.0.0'
```

Review `Summary.html`, `Regression-Checks.csv`, `Capture-Coverage.csv`, `Shortcut-Reconciliation.csv`, `Likely-Migration-Gaps`, and `Manual-Secure-Actions.csv`.

8. Rollback

Always preview with the exact backup path printed by the repair script:

```
.\Rollback-PCMigrationRepair-v4.0.0.ps1 `
  -BackupPath 'C:\Users\USER\Desktop\PCMigration-v4.0.0-Backup-YYYYMMDD-HHMMSS'`
```

Apply rollback:

```
.\Rollback-PCMigrationRepair-v4.0.0.ps1 `
  -BackupPath 'C:\Users\USER\Desktop\PCMigration-v4.0.0-Backup-YYYYMMDD-HHMMSS' `
  -Apply
```

If a registry subtree did not exist before repair and the preview confirms it is safe to remove:

```
.\Rollback-PCMigrationRepair-v4.0.0.ps1 `
  -BackupPath 'C:\Users\USER\Desktop\PCMigration-v4.0.0-Backup-YYYYMMDD-HHMMSS' `
  -RemoveNewRegistryKeys `
  -Apply
```

Package installations, Windows features, and Windows capabilities are recorded but are not automatically uninstalled during rollback.

Repair methods and reversibility

Method	Automatic backup/rollback
CopyFile / CopyShortcut	Existing destination file is backed up; a newly created file can be removed by rollback.
ImportRegistry	Existing narrow subtree is exported; rollback deletes the changed subtree before exact re-import.
Set24HourClock	International registry subtree is backed up.
SetTimeZone	Previous time-zone ID is recorded; review rollback results.
WingetInstall	Recorded as non-reversible; uninstall is manual.
EnableFeature / AddCapability	Recorded as non-reversible; disabling/removal is manual.

9. Unicode and special-path handling

Windows permits many non-ASCII filename characters. Version 4.0.0 uses Unicode-aware .NET filesystem APIs, UTF-8 CSV/JSON/text output, literal path operations, and source-manifest hashing.

The filename character μ (U+00B5 MICRO SIGN) survives capture, reporting, hashing, payload copying, repair authorization, and restoration. It remains distinct from μ (U+03BC GREEK SMALL LETTER MU).

Comparison uses Unicode Form C to stabilize canonically equivalent text. It deliberately does not use compatibility normalization, which could collapse distinct legal characters. File operations continue to use the original captured literal path.

Case	Behavior
μ versus μ	Preserved as distinct characters; regression-tested.
Composed/decomposed accents	Comparison text uses Unicode Form C; literal source/destination paths remain unchanged.
Spaces and punctuation	Preserved in paths; shell interpolation is avoided through LiteralPath/.NET APIs where applicable.
Parent traversal	Rejected for repair artifacts and shortcut destinations.
Reparsing points	Skipped to avoid loops and unintended volume traversal.
Different user names	Known roots are tokenized, such as %APPDATA% and %USERPROFILE%.

Application caveat

The toolkit can preserve and compare a Unicode path, but it cannot make a legacy third-party application Unicode-correct. Review any application that independently rejects, rewrites, or normalizes such paths.

Path safety boundaries

- Source artifacts must be capture-relative and remain inside the validated capture.
- Destination file paths must resolve through an approved root token.
- Shortcut destinations must resolve through a supported known-folder root plus a traversal-free relative path.
- Broad registry ` .hiv` files are not included in repair authorization maps.

10. Capability matrix

Category	Automatic or selective support	Manual or excluded
Applications	AppX/MSIX, WinGet, uninstall, Start, shortcut, and portable evidence; exact authorized WinGet installs.	Unknown package identity, license transfer, vendor-specific installation.
Settings files	Configuration/script inventory and screened source payload copy.	Unknown databases, protected stores, incompatible schemas.
Registry	Value-level comparison; narrow approved .reg payloads; optional forensic snapshots.	Whole-hive restore; protected credential/authentication branches.
Shortcuts	Full reconciliation, semantic comparison, approved copy with backup.	Pinned shell databases and unsupported taskbar/Start internals.
Windows state	Selected clock/time zone/settings methods; broad inventory.	Policy/domain/hardware-specific decisions.
Integration	Evidence for services, tasks, printers, VPN/Wi-Fi, firewall, shares, ODBC, certificates.	Generic automatic recreation of high-impact integration state.
Credentials/security	Checklist and presence metadata only.	Secrets, DPAPI, Hello, passkeys, private keys, tokens, antivirus databases.
Development tools	Tool/version, Git, WSL, VS Code, PowerShell evidence.	Toolchain-specific license/authentication and environment reconstruction.

Protected and volatile exclusions

Defender and Kaspersky runtime databases, Windows Search indexes, notification/activity stores, diagnostics databases, thumbnail/icon caches, browser session databases, credential stores, and other protected or rebuildable state are excluded before generic copying. Security products are not disabled to force access.

Incomplete coverage

A failed or partial destination collector disables missing-item conclusions for that category. This prevents an access failure from becoming a false migration gap. Review `Capture-Coverage.csv` before every repair plan.

11. Disclaimers and limitations

- This toolkit is provided as-is under the MIT License, without warranty.
- It is not affiliated with Microsoft, Laplink, PCmover, WinGet package publishers, or any application vendor.
- It does not guarantee completeness, compatibility, or identical behavior between machines.
- A hash or state difference proves a difference, not that the source should overwrite the destination.
- Windows build, architecture, hardware, domain, policy, security-product, and application-version differences may make source state inappropriate.
- Captures and reports can expose software inventory, user paths, configuration, network metadata, and other sensitive information.
- Optional registry binary snapshots are highly sensitive and should never be published or committed to Git.
- Credentials, private keys, authentication sessions, and protected databases require vendor-supported or manual migration.
- Maintain independent image/file backups and keep the source operational until destination verification is complete.

GitHub publication checklist

- Publish the clean source package, README, PDF guide, changelog, security policy, license, and checksum manifest.
- Do not publish any source/destination capture, report, repair plan, backup folder, registry hive, or machine-specific diagnostic.
- Create a release ZIP from the validated package directory and publish its SHA-256 digest.
- Document that v4.0.0 requires fresh schema 4.0 captures.
- Use private security reporting for vulnerabilities; require sanitized reproductions.

Final operational rule

Capture, compare, inspect, install applications, recapture, regenerate the plan, approve individual actions, preview, apply, sign out, verify, and retain rollback evidence. Never skip directly from first comparison to bulk approval.

End of guide